HCLTech Career Shaper Programme — III Year Industry Project

Comprehensive Project Brief & Technical Specification | Academic Session 2025–26

Project Specification Field	Candidate & Project Details
Project Title	**Agentic Web Scraper and Market Intelligence Reporter**
Project Identification No.	**Project 81**
Specialization Domain	**Agentic AI / Python** (Intelligent Autonomous Systems Track)
Learner / Candidate Name	**Adarsh Dubey**
University Roll Number	**2400320100061**
Academic Institution	**ABES Engineering College, Ghaziabad** (Dept. of Computer Science & Engineering)
Industry Project Mentor	**Radhika Singhal** (HCLTech Career Shaper Industry Mentor)
Core Technology Stack	Python 3.11+, LangChain, LangGraph, Google Gemini 2.5 Flash, DuckDuckGo, Trafilatura, Pydantic v2, Streamlit, Plotly
Difficulty / Project Duration	Intermediate to Advanced Industry Capstone | **4 Weeks** (Sprint-based)
GitHub Repository	https://github.com/adarsh-dubey-gthb/hcl-marketanalyzer.git
Submission Format & Scope	Individual Comprehensive Project Deliverable (1 Learner) | Academic Year 2025–26

1. Executive Summary & Problem Formulation

Modern business strategists, market analysts, and corporate intelligence officers face intense operational friction when monitoring dynamic competitor movements, pricing fluctuations, technological patents, and enterprise product rollouts. Conventional market research is notoriously fragmented, manual, and cost-prohibitive. Furthermore, traditional heuristic web scrapers (e.g., rigid CSS/XPath scripts) fail catastrophically when confronted with modern dynamic DOMs, client-side rendering (React/Next.js), anti-bot challenges, and frequent structural website updates.

This project bridges the intelligence gap by engineering an **Autonomous Agentic Web Scraper and Market Intelligence Reporter**. Powered by state-of-the-art LLMs (Google Gemini 2.5 Flash) orchestrated via LangGraph ReAct state machines, the system operates as an independent market researcher: it receives high-level research directives, dynamically formulates targeted search strategies, crawls competitor domains with polite rate-limiting, extracts verified structured data, validates entities against strict Pydantic v2 models, and synthesizes executive-ready Markdown, HTML, and JSON dossiers alongside an interactive Streamlit visual analytics console.

2. Business & Engineering Objectives (Quantitative KPIs)

Performance Dimension	Target KPI	Operational Definition & Verification Method
Extraction Accuracy & Precision	≥ **94.0%**	Verified entity extraction accuracy audited against target corporate source pages with zero PII leakage.
Query-to-Report Response Latency	≤ **5.0 s**	Average wall-clock response time for multi-source intelligence retrieval and automated report synthesis.
Scraper Uptime & Recovery Yield	≥ **98.0%**	Successful job execution rate across diverse domains using Trafilatura fallbacks and exponential backoff.
Manual Research Time Reduction	≥ **85.0%**	Measurable operational time savings compared to human analyst manual web search and manual synthesis.
Data Schema Integrity	**100.0%**	Zero validation errors; 100% adherence to Pydantic v2 models with zero null primary keys.

3. Technical Architecture & Component Breakdown

• **Agentic Orchestration Subsystem (LangGraph ReAct Loop):** Implements a stateful directed acyclic graph (DAG) coordinating four specialized agent roles: *Research Planner* (decomposes the objective into focused queries), *Web Navigator & Tool Operator* (invokes search and crawl tools), *Observation Critic* (validates data relevance and truthfulness), and *Report Synthesizer* (assembles structured intelligence). Operates on continuous Thought → Action → Observation feedback cycles.

• **Polite Web Discovery & Harvesting Engine:** Combines DuckDuckGo Search API for live web discovery with Trafilatura for primary main-text extraction. Enforces strict compliance with robots.txt, polite request throttling (1.5s delay between hits), automated User-Agent rotation, and graceful HTTP 429/403 backoff handling.

• **DOM Normalization & Data Cleansing:** Strips boilerplate HTML, inline scripts, CSS, navigation menus, and third-party advertising iframe noise, transforming unstructured web pages into clean, token-efficient semantic Markdown.

• **Pydantic v2 Schema Enforcement:** Validates every extracted entity against rigid schemas (MarketIntelligenceReport, CompetitorEntity, FinancialMetrics, RiskFactor) to guarantee structural immutability and eliminate LLM hallucinations.

• **Multi-Format Intelligence Exporter:** Produces three synchronized outputs: clean GitHub-flavored Markdown, standalone self-contained executive HTML with embedded typography, and raw machine-readable JSON for enterprise downstream ETL ingestion.

• **Interactive Streamlit Visual Dashboard:** Provides an intuitive web UI supporting user parameter configuration (focus pillars, regional bounds, exclusions), real-time live execution logs, Plotly visual charts, and instant report downloads.

4. Key Deliverables & Final Submission Checklist (Due at Week 4)

Status	Deliverable Name & Format	Specification & Scope Description
[X]	**Working Application Prototype** market_analyzer/	Fully operational Python agent application with Streamlit interactive UI, LangGraph ReAct engine, and multi-format reporting.
[X]	**Project Synopsis Document** Project_Synopsis_Adarsh_Dubey_...	Completed, instruction-free formal synopsis in PDF and DOCX formats with candidate details and 300 DPI embedded diagrams.
[X]	**Benchmark Telemetry Dataset** dataset.csv (5,000 rows)	5,000 production-grade scrape logs across 51 real corporate domains in 10 industry sectors with 0 missing values.
[X]	**Data Dictionary Specification** data_dictionary.csv, .pdf, .docx	Exhaustive column specification documenting data types, descriptions, constraints, nullability, and sample values.
[X]	**Dataset Report & Telemetry Profile** Dataset_Report.pdf & .docx	2-page analytical report featuring executive telemetry, 10-sector domain breakdown, 15 benchmark sample rows, and statistical EDA.
[X]	**Comprehensive Project Report** Technical Documentation (≥ 8 pages)	Individual comprehensive technical report detailing system design, algorithmic workflows, testing regimes, and results.
[X]	**EDA Script / Notebook with 3+ Plots** eda_telemetry.ipynb / Script	Exploratory Data Analysis script demonstrating crawl yields, duration distributions, and error anomaly taxonomies.
[X]	**Video Walkthrough Demonstration** MP4 Video (1–2 minutes) / GitHub Repo	Concise screencast showing live competitive intelligence extraction, agent tool-use execution, and UI report generation.
[X]	**Signed Ethical Compliance Declaration** Formal Written Endorsement	Verified affirmation of DPDP Act 2023 compliance, robots.txt adherence, server politeness, and zero PII harvesting.

5. Detailed 4-Week Sprint Implementation Roadmap

Sprint / Week	Focus Area & Scope	Engineering Tasks & Milestones	Mentor Checkpoint & Review
Week 1 (Oct 06–12)	Problem Understanding, Toolchain Setup & EDA	• Configure virtual environment (Python 3.11+, LangChain, Gemini API). • Perform statistical EDA on dataset.csv (5,000 scrape sessions). • Draft schema and publish data_dictionary.csv.	**Checkpoint 1:** Mentor signs off on problem formulation, Data Dictionary, and exploratory findings.
Week 2 (Oct 13–19)	Architecture Design & Core Agentic Engine (Sprint 1)	• Implement LangGraph ReAct state machine. • Integrate DuckDuckGo Search and Trafilatura content tools. • Construct Pydantic v2 schemas for entity extraction. • Implement polite crawler with robots.txt parser and 1.5s delay.	**Checkpoint 2:** Live CLI demo to mentor verifying agentic crawl and structured entity extraction.
Week 3 (Oct 20–26)	Multi-Domain Benchmarking, Streamlit UI & Reports (Sprint 2)	• Build Streamlit dashboard with custom research constraints. • Implement Markdown, executive HTML, and JSON report exporters. • Benchmark against 51 corporate domains across 10 sectors. • Record performance metrics (latency, precision, recovery).	**Checkpoint 3:** End-to-end dashboard walkthrough; mentor validates ≥94% extraction accuracy.
Week 4 (Oct 27–Nov 02)	Comprehensive Validation, Documentation & Final Expo	• Compile Project Synopsis, Project Brief, and Final Report (≥8 pages). • Record 2-minute video demonstration and clean repository. • Execute ethical audit and obtain mentor endorsements. • Complete final submission and project expo defense.	**Checkpoint 4:** Formal evaluation by HCLTech panel; final submission of all 5 project deliverables.

6. Production Tech Stack & Component Specifications

Component Layer	Technologies & Libraries	Architectural Responsibility in Project 81
Programming Language	Python 3.11+	Core runtime environment with asynchronous I/O and strict type annotations.
Agentic Orchestration	LangChain v0.3+, LangGraph v0.2+	Stateful multi-actor graph execution, ReAct cycles, tool dispatching, and agent memory.
Foundation LLM	Google Gemini 2.5 Flash / Gemini Pro	High-speed reasoning, multi-query generation, structured extraction, and synthesis.
Web Discovery & Search	DuckDuckGo Search API, Urllib	Autonomous discovery of competitor URLs, corporate press releases, and SEC filings.
HTML Extraction & Parsing	Trafilatura, BeautifulSoup4, LXML	Resilient text discovery, boilerplate removal, and DOM normalization across messy websites.
Validation & Schema	Pydantic v2.8+, Dataclasses	Strict validation of extracted corporate entities, ensuring 100% schema consistency.
User Interface & Charts	Streamlit v1.38+, Plotly Express	Interactive analyst console, live execution streaming, and visual market share plots.
Document Generation	ReportLab 4.x, python-docx, PyMuPDF	Automated compilation of submission-ready PDF and Word deliverables with embedded diagrams.

7. Benchmark Dataset Specifications & Schema Summary

Included within this project repository is the official benchmark dataset (dataset.csv), comprising **5,000 real-world scrape telemetry sessions** evaluated across **51 corporate domains** in 10 industry sectors (IT Services, Cloud SaaS, Fintech, Cybersecurity, Retail, Health, Logistics, AI, EdTech, Telecom). The dataset demonstrates an empirical 83.5% success rate, 315.8-second average scrape duration under polite rate-limiting, and 100% data completeness (0 null values).

8. Authoritative Academic & Industrial References

1. **LangChain & LangGraph Official Documentation:** Building Autonomous Multi-Actor LLM Applications with State Graphs (https://python.langchain.com/)

2. **Google DeepMind Gemini API Documentation:** Structured Outputs, ReAct Agent Patterns, and Tool Calling Architecture (https://ai.google.dev/)

3. **Yao, S., Zhao, J., Yu, D., et al. (2022):** *ReAct: Synergizing Reasoning and Acting in Language Models*. International Conference on Learning Representations (ICLR).

4. **Barbaresi, A. (2021):** *Trafilatura: A Web Scraping Library and Command-Line Tool for Text Discovery and Extraction*. Proceedings of the 59th Annual Meeting of the ACL.

5. **Digital Personal Data Protection (DPDP) Act 2023:** Ministry of Electronics and Information Technology, Government of India (Official Gazette of India).

6. **HCLTech Career Shaper Programme Curriculum & Capstone Guidelines 2025–26:** Provided by Programme Coordination Unit.

9. Ethical, Legal & Regulatory Compliance Framework

ETHICAL SCRAPING DIRECTIVE & DPDP ACT COMPLIANCE MANDATE:
This project strictly enforces ethical web scraping protocols and full compliance with India's **Digital Personal Data Protection (DPDP) Act 2023**.
1. **Robots.txt & Terms of Service:** The autonomous agent parses and honors robots.txt directives and disallow rules before initiating crawls.
2. **Server Politeness & Rate Throttling:** Enforces a mandatory minimum delay of 1.5 seconds between consecutive requests. Parallel flooding and denial-of-service patterns are strictly prohibited.
3. **Zero Personal Data / PII:** The crawler is strictly restricted to publicly accessible, non-authenticated corporate market data. No individual user profiles, contact lists, or credentials are collected or retained.
4. **Transparent Identification:** All outbound HTTP requests carry a descriptive User-Agent header identifying the agent as an educational market intelligence research tool.

10. Formal Project Endorsement & Signatures

Learner / Candidate Endorsement: Name: Adarsh Dubey Roll Number: 2400320100061 Institution: ABES Engineering College, Ghaziabad Date: October 30, 2026	**Industry Project Mentor Endorsement:** Name: Radhika Singhal Designation: Industry Project Mentor Organization: HCLTech Career Shaper Programme Approval: **[APPROVED FOR SUBMISSION]**